

Guanine Exchange Factors Show Existence of Transient Pockets on Rac1 Surface

Elvis A. F. Martis,[†] Vincent Sauzeau,[‡] and Stéphane Téletchéa^{*,†}

[†]*US2B, Nantes Université, CNRS, UMR 6286, F-44000 Nantes, FRANCE*

[‡]*UMR_S 1087 / UMR_C 6291 l'Unité de Recherche de l'Institut du Thorax, F-44000 Nantes, FRANCE*

E-mail: stephane.teletchea@univ-nantes.fr

Abstract

Rac1 (Ras-related C3 botulinum toxin substrate 1) is a molecular switch that catalyses the hydrolysis of GTP to GDP. Owing to the slow nature of GTP hydrolysis and even slower dissociation of GDP, Rac1 must be assisted by Guanine Exchange Factors (GEFs) and GTPase-activating proteins (GAPs). As a result, GEFs and GAPs control the switching mechanism in Rac1. This regulatory control mechanism by several GEFs and GAPs provides a means to modulate the Rac1 activity in several diseases in which it has been implicated. The work described here attempts to understand various dynamical aspects of Rac1 in the presence of GTP, GDP, and guanine exchange factors (GEFs) to locate transient binding sites on the Rac1 surface. We use co-solvents as chemical probes to accelerate pocket-opening due to induced fit mechanism. We observe four transient sites on the RAC1 surface. Two of which form the docking site for GEFs. One is observed where the DHR domain containing GEFs from the DOCK family bind. And another is located at the site where the ELMO1 protein interacts with RAC1. In the absence of GEFs, the pocket volumes are too small to be druggable, indicating their transient nature. Using the pocket volume fluctuations, we evaluate their significance for targeting small molecules. Furthermore, we use principal component analysis to understand

the protein dynamics that are essential for the opening and closing of these transient pockets.

Introduction

Small GTPases, a ubiquitous class of proteins, play pivotal roles in various cellular processes. Their functions include signal transduction, vesicle trafficking, cytoskeletal dynamics, and cell division. Ras GTPases regulate cell proliferation, differentiation, and survival. Rho GTPases govern cytoskeletal dynamics and cell migration. Their activity is regulated through a molecular switch mechanism by guanine nucleotide exchange factors (GEFs), GTPase-activating proteins (GAPs), and guanine nucleotide dissociation inhibitors (GDIs).¹ These molecular switches cycle between OFF (GDP-bound inactive) and ON (GTP-bound active) states. This switching mechanism controls various processes like cell growth and differentiation to vesicular and nuclear transport.^{1,2}

Small GTPases are proteins that possess conserved domains across several families; made up of six β -sheets surrounded by 5 α -helices. All domains are conserved such that they carry a guanine nucleoside binding site to recognise the guanine base and phosphate groups and the magnesium ion.^{1,3} The switch regions regulate the traffic to the guanine nucleoside binding site. It is also well established that GTPases alone cannot hydrolyse GTP effectively. Therefore, GEFs and GAPs play a significant role in assisting the switching mechanism of GTPases. They control the dynamics of the switch regions that facilitate either the hydrolysis of GTP or the dissociation of GDP. GDP:Mg²⁺ dissociates from GTPases very slowly, and therefore, GEFs assist in exchanging GDP for GTP. One of the accepted mechanisms of GEFs in the dissociation of GDP is by electrostatic repulsion of the magnesium ion that forms a strong interaction with GTPases and phosphate groups of GDP. The loss of a magnesium ion weakens the affinity of GDP, which eventually dissociates. Alternatively, some GEFs can rearrange the switch I or II conformations such that GDP's affinity is reduced, facilitating its dissociation. However, GEFs can also combine the two mechanisms to facilitate GDP dissociation.^{4,5}

Furthermore, GAPs stabilise the transition state for GTP:Mg²⁺:GTPases and facili-

tate the hydrolysis of the γ -phosphate group from GTP. Taken together, the switching mechanism can be seen as GEFs assist in the GTPase activation, while GAPs assist in GTP hydrolysis. Following GTP hydrolysis, the inactive GTPases prevent further signal transduction.⁶ In some classes of GTPases, the active and inactive states are conjugated with GTPases being shuttled between the cell membrane and cytosol. Such GTPases undergo post-translational modifications to carry long-chain fatty acids at their C-terminal to assist bilayer insertion. GDIs^{7,8} sequester the lipid component to prevent membrane insertion of GTPases in the cytosol. This coordinated action of GTPases, GEFs, GAPs, and GDIs is essential for various cellular processes. Dysregulation of these components contributes to pathologies like cancer, neurodegeneration, and inflammatory disorders. For instance, mutations in Ras and its regulators are implicated in oncogenesis, highlighting the therapeutic potential of targeting GTPase signalling pathways.¹ Such mutations are known to disrupt the equilibrium between the active and inactive states of GTPases. It has also been observed that some mutations enhance the affinity of GEFs or GAPs that trap the small GTPases either in an activated state or an inactivated state. Small molecule inhibitors targeting GTPase activity or disruptors of GEF/GAP interactions⁹ are under development for cancer therapy. However, there are several challenges in targeting GTPases and/or GTPases-GAP/GEF complexes. Firstly, targeting the guanine binding site is challenging owing to the large intracellular concentration of GTP and high affinity.¹⁰ In such cases, finding alternate sites can avoid such competition with endogenous substrates. GTPase hyper-activation or inactivation can be corrected by disrupting the protein-protein interaction (PPI) between GTPases and GEFs/GAPs by targeting the regulatory protein docking sites.

Methods

- List of PDBS used. All modelled systems are described in Table 1.

Table 1: Detailed system studies

Initial PDB	Rac1 Modulator/LIGAND	Comments
1HE1 ¹¹	GDP: Mg^{2+}	Inactive Rac1
3TH5 ¹²	GTP: Mg^{2+}	Active Rac1
4YON ¹³	PREX1	DBL family member
1FOE ¹⁴	TIAM1	DBL family member
2VRW ¹⁵	VAV1	DBL family member
2YIN ¹⁶	DOCK2	DOCK family member

Preparation of systems for molecular dynamics simulation:

The coordinates for the protein-ligand complexes were retrieved from the Protein Data Bank (PDB); the PDB entries used in this study are listed in **Table 1**. The X-ray structures were imported and processed in the UCSF Chimera v1.17.¹⁷ Any missing loops were built using Modeller¹⁸ with UCSF Chimera v1.17 as the GUI.^{19,20}

Generating AMBER-compatible parameters:

All systems were prepared for MD simulations with the Leap module in AMBERtools23. Protein atoms were atom-typed according to the AMBER14 (ff14SB) force field.²¹ Amber-compatible parameters for all nucleotides were obtained from the Manchester AMBER database. For the VAV1 zinc finger region, the parameters of the Zinc atom, coordinating cysteines and histidine residues, the force field parameters were obtained from the ZAFF²² force field library.

Molecular dynamics simulations:

All complexes were solvated with TIP3P²³ waters enclosed in a box with dimensions extending 10Å beyond any protein atom, and an appropriate number of neutralising ions were added to maintain the system’s neutrality. The complexes were relaxed using three cycles of minimisation of 50000 steps each. The first 5000 steps of minimisation adopted the steepest descent algorithm, and the remaining 45000 steps used the conjugate gradient algorithm. The cut-off for evaluating non-bonded interactions was placed at 8.0Å for the minimisation phase. The first cycle of minimization was performed with 25 kcal/(molÅ²

restraining force on the solute atoms, this force constant was reduced to 5 kcal/(molÅ²) in the next cycle, and the final cycle was performed without any restraining forces on either the solute or the solvent atoms. The system was heated in 4 stages (see **Table 2**) from 0 K to 300 K for 10 ns with 50 kcal/(molÅ²) restraining force on the solute atoms using the Langevin dynamics, with a collisional frequency of 2 ps⁻¹. The next step employed the NPT ensemble for density equilibration for another 2 cycles of 10 ns each, with the restraining force on the solute atoms gradually decreasing from 5 to 0 kcal/(molÅ²). In the equilibration phase, the cut-off for evaluating non-bonded interactions was placed at 8 Å. In the production phase of the MD simulations, all systems were simulated for 100 ns in triplicate, amounting to a 300 ns trajectory for each system. The long-range electrostatic interactions were calculated using the Particle Mesh Ewald (PME) summation method.²⁴ The cut-off for evaluating non-bonded interactions was placed at 9.0 Å for the production phase. All the computations were done using the CUDA-enabled PMEMD code²⁵⁻²⁷ in AMBER22.^{28,29} All simulations were performed with the SHAKE algorithm³⁰ to constrain bonds to hydrogen atoms, with the tolerance set to 0.00001 Å. A 2 fs integration time step was used to solve Newton’s equations of motion. For analysis, the conformations were saved every 100 ps, amounting to 1000 snapshots per trajectory (3000 per system).

Table 2: Details of stages of Heating using NVT Ensemble

Stage	Temperature range (K)	Time (ns)
Stage 1	0 to 100	0 to 2.5
Stage 2	100 to 200	2.5 to 5
Stage 3	200 to 300	5 to 7.5
Stage 4	300 to 300	7.5 to 10

Measuring the structural dynamics of Rac1:

Analysis of the MD trajectory was performed with the cpptraj³¹ module in AMBER-Tools23. Before the analysis, all solvent atoms and neutralising ions were stripped off, and the protein conformations were aligned to their respective X-ray conformation. The mass-weighted C α root-mean-square deviation (RMSD) for the protein was computed using the X-ray conformation as the reference. The root-mean-square fluctuations (RMSF)

per residue for the backbone and side-chain atoms were computed to understand the domain-specific fluctuations. Furthermore, to understand the specific dynamical changes in Rac1 when bound to different binding partners, we extracted the essential dynamical properties using Principal Component Analysis (PCA). The Variance-Covariance matrix for C α atoms was extracted in the cpptraj³¹ module in AMBERTools23. We extracted and analysed 10 principal components; however, the first two principal components explained $< 50\%$ variance in the data. To visualise the essential dynamics, the vectors were projected on the C α atoms and visualised in VMD 1.9.3.³²

Identification and measurement of alternate pockets on Rac1:

POVME2^{33,34} was employed for identifying various pockets on the Rac1. To be able to incorporate various experimental conformations of Rac1, we employed Rac1 with different binding partners (see **Table 1**). To identify the pockets, the pocketID module on POVME2 was used on the aforementioned Rac1 structures. The parameters used for pocket identification are shown in **Table 3**. Following the pocket identification, we monitored the pocket volume changes of these pockets throughout MD trajectories. It has been observed that the pocket might be a good candidate for druggability if its volume range^{35–37} was found to lie between 610Å³ to 930Å³.

Table 3: Parameters used for pocket identification

Parameter	Value
pocket detection resolution	4.0 Å
pocket measuring resolution	1.0 Å
clashing cutoff	3.5 Å
number of neighbors	4
number of spheres	5
sphere padding	5.0 Å

Co-solvent MD simulations:

Co-solvent MD simulations are very useful to keep the transient pockets open for a longer timescale in MD simulations to better understand pocket characteristics. We identified

transient pockets in response to GEF binding, which were otherwise not observed or very hard to locate. To ensure that these transient pockets can remain open sufficiently long to be druggable, we used co-solvent MD simulations. For this purpose, we used 1M concentration of isopropanol, phenol and toluene. The choice of these solvents mimics drug-like fragments with decreasing water solubility. In this case isopropanol ($\log P = 0.05$) > phenol ($\log P = 1.39$) > toluene ($\log P = 2.56$). The simulation box with Rac1 protein (PDB: 3TH5, GTP and Magnesium ions were deleted) and 0.15M of NaCl was prepared using packmol.³⁸ The protocol for MD simulations is the same as mentioned above. GAFF2 parameters were assigned to the organic solvents. For co-solvent MD simulations, we performed triplicate MD simulations for $1\mu s$, amounting to a total sampling time of $3\mu s$ per solvent system. For the baseline calculations, we also prepared two systems without adding co-solvents. One is the RAC1 complexed with GTP and Magnesium ions, that mimics an active like conformation of RAC1. The apo system was created by deleting GTP and Magnesium ions from the active like system described previously. These systems would enable use to understand if any of the transient pockets can spontaneously open without the influence of the co-solvents and any ligand in the nucleotide pocket.

Quantifying the GEF binding hotspots on Rac1 surface:

The free energy of binding was computed using the equation (Eq-1) on the conformations saved from the MD simulations using the mm-pbsa python script³⁹ available in AMBERTools23;

$$\Delta G_{\text{binding}} = \Delta E_{\text{MM}} + \Delta H_{\text{GBSA}} - T\Delta S \quad (1)$$

Where ΔE_{MM} is the interaction energy between the binding partners computed in the gas phase using a molecular mechanics method. The solvation-free energy or the enthalpy of hydration ΔH_{GBSA} was computed using the Generalised-Born Surface Area (GBSA) method, and the entropic component of binding $T\Delta S$ was not computed due to the huge computation cost. The internal dielectric constant for GBSA calculations was set to 1, and the solvent dielectric constant was fixed at 80. For computing the

free energy of binding using the MM-GBSA method, the OBC’s GB model⁴⁰ ($igb = 5$) was used. The mbondi3 radii set was employed to define the PB radii for all solute atoms during the preparation of the topology and parameter files in the leap module in AMBERTools23. The ionic strength for GBSA calculations was maintained at 150 mM. All energy components, in an implicit solvent, were calculated using a large cut-off of 999 Å. The surface area for the non-polar solvation term was computed using the linear combination of pairwise overlaps (LCPO) method.⁴¹ To identify the critical amino acid residues that are critical for the protein-protein interactions between Rac1 and GEFs, we computed the per-residue decomposition of the binding affinity value. This will also enable us to estimate the contribution from the residues lining the newly identified transient pocket and their energetic contribution to GEF binding.

Results and discussion

Rac1 plays a crucial role in regulating cellular processes owing to its switching mechanism that recruits several proteins upstream and downstream. The GTP-bound state is called the *Active* state and the GDP-bound state is called the *Inactive* state. The energy obtained from the GTP hydrolysis is used to undergo conformational changes that are partially responsible for the switching mechanism. The rest of the conformational switching is assisted by GTPase-activating proteins (GAPs) or Guanine Exchange Factors (GEFs). GTP hydrolysis is a very slow process and must be assisted by GAPs to ensure its completion in biologically relevant time scales. Owing to the picomolar affinity of the GDP to Rac1, GEFs are necessary for GDP dissociation. Therefore, GEFs are known as exchange factors that facilitate GDP dissociation and GTP association. There are several GEFs known, and specific GEFs are active on specific GTPases. PREX1, TIAM1, VAV1 and DOCK2 are Rac1-specific GEFs. They regulate many cellular processes and, therefore, are implicated in many disease states. This makes them attractive targets to restore their abnormalities. Structurally, PREX1, TIAM1 and VAV1 are DH/PH domain-containing proteins, while DOCK2 contains a DHR domain. Irrespective of their structural diver-

sity, GEFs bind to the switch I and/or II regions. The nucleotide-binding site is well conserved among the GTPase family. This poses a significant challenge to selectively target small GTPs. Furthermore, the molar concentrations of GTPs and picomolar binding affinity make it more difficult for the competing small molecules. The switching mechanism offers a conformational landscape that must be studied to locate new binding sites. The association of the GEFs show the existence of alternative binding sites that might transiently open to allow docking of the GEFs. Therefore, we embarked to understand the complete dynamical landscape of Rac1 protein bound to either nucleotides (GDP or GTP) or GEFs (PREX1, TIAM1, VAV1 or DOCK2). We expect to locate alternate binding pockets that may improve the druggability of the Rac1 protein. We also attempt to understand if different binding partners show differences in the pockets we identify, which can be exploited to selectively target Rac1 implicated in different signalling pathways. We hypothesise that the selectivity can be measured as a function of the pocket volume changes. We also attempt to establish if there exists any correlation in the volume changes amongst other pockets or with the nucleotide binding site. The latter can be believed to influence the nucleotide traffic and eventually control the switching mechanism of Rac1.

GEF docking site houses transient pockets on the Rac1 surface

We employed the POVME2 tool to identify and measure the pocket volume changes associated with each pocket. To locate new pockets, we simulated Rac1 under different conditions (See **Table 1**). Before identifying the binding pockets, all binding partners, including nucleotides, were deleted from the trajectory. This allowed us to focus entirely on the conformations of Rac1 to identify new pockets. The pockets were identified using the starting conformation of Rac1 X-ray structure obtained under different conditions (See **Table 1**). The pocket volume changes were monitored throughout the course of the trajectory. To understand the volume changes of the nucleotide binding site in response to various Rac1 binding partners(See **Table 1**), we measured and compared the volume changes under different binding conditions. The pocket volume fluctuations are

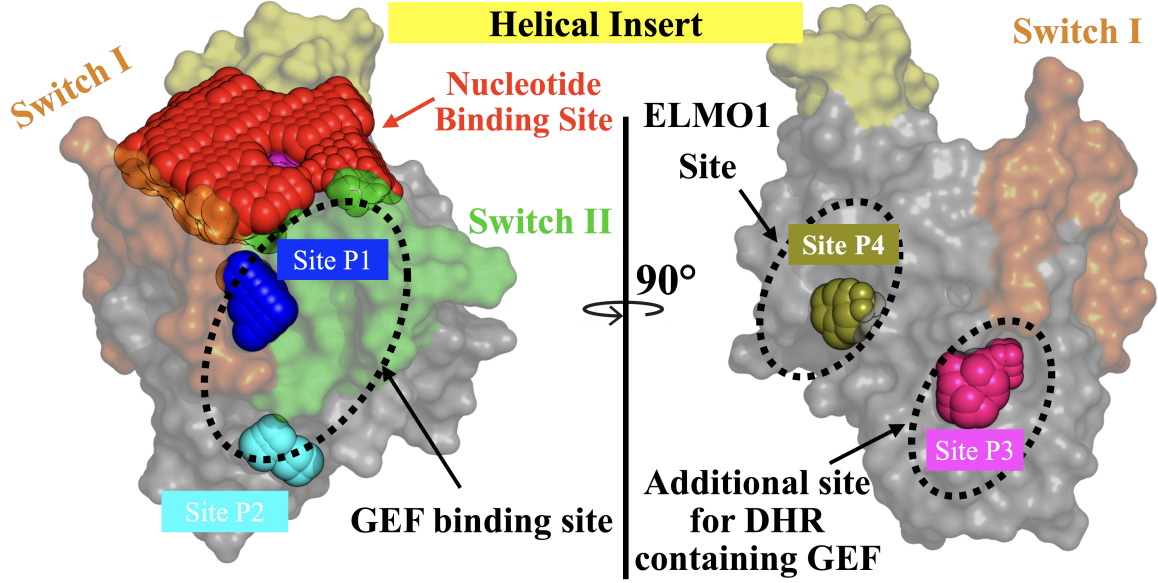


Figure 1: The Surface of Rac1 with all plausible binding sites.

restricted under $3000\text{\AA}^3$ in the presence of GTP and GDP along with the coordinated magnesium ion (**Figure 2B**). This shows closing of the nucleotide pocket to initiate the GTP hydrolysis (GTP-bound) or a state would be likely after the GTP hydrolysis (GDP). We also observed that switch region dynamics was reduced significantly indicating stabilising effect of with Mg^{2+} that interacts with the switch regions (**Figure 5D and 5E**). The switch region conformation is also found to be reduced by GEF binding (**Figure 5D and 5E**), suggesting their role in controlling the nucleotide traffic by modulating the switching regions. We tested this hypothesis by simulating a nucleotide-free and GEF-free form of Rac1 (delGDP and delGTP in **Figure 2B**) and found that the mean pocket volume and fluctuations showed a greater tendency to increase, however, only marginally, as maximum volume was still under $4000\text{\AA}^3$. We speculate that the pocket could have opened even more had the simulation time been longer. Next, we measured the pocket volume and the fluctuations in the presence of GEF in the nucleotide-free state. The Volume fluctuations are higher than in the nucleotide-free state without GEFs. However, the volume changes showed significant changes with different GEFs. For example, PREX1 and DOCK2 showed volume changes that are similar in magnitude to nucleotide-free state without GEFs (See delGDP and delGTP in **Figure 2B**). TIAM1

and VAV1 showed significantly enlarged nucleotide pockets with mean volumes above 3200Å³. We hypothesise that such a difference in pocket volume changes in the presence of different GEFs suggests that the X-ray structure has perhaps captured the Rac1-GEF complexes either just after GDP dissociation (PREX1 and DOCK2) or in a state that was pre-organised for GTP: Mg^{2+} association (TIAM1 and VAV1). The changes in the pocket volumes for the nucleotide binding site can be attributed to the changes in the switch region dynamics that govern the traffic in the nucleotide site. The P-loop dynamics also govern the pocket changes that are evident from the GEF- and nucleotide-free states, where the P-loop is more flexible (see figure 5C).

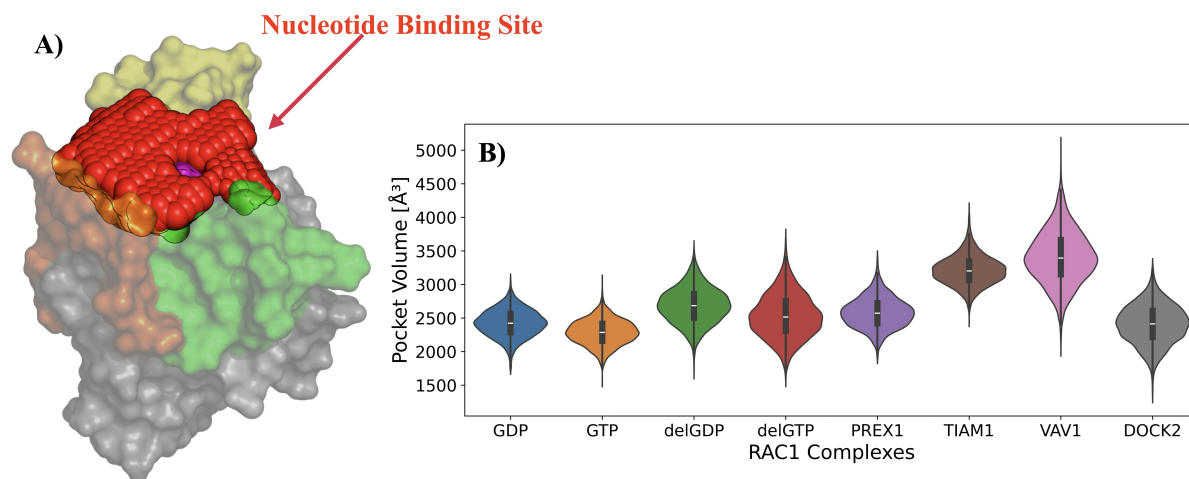


Figure 2: **A)** Shows the surface model of Rac1 protein with coloured spears illustrating the binding site identified with POVME2. The helical insert is highlighted using the yellow colour. The green region indicates the switch II, and the orange region indicates the switch I region. The magenta region slightly visible through the red spears indicates the P-loop. The red spears occupy the nucleotide binding site that is lined with helical insert, switch regions, and the P-loop forms the phosphate binding region. Blue spears occupy the site P1, and the cyan spears occupy the site P2. Sites P3 and P4 are visible by rotating the Rac1 by 90°. The salmon pink spears occupy site P3 while the olive green spears occupy site P4 **B)** The violin plots illustrate the distribution of the pocket volumes for each of the nucleotide binding sites. The changes in the pocket volumes are monitored when Rac1 is complexed with either GEFs, GDP, GTP or in the *apo*-state.

We identified four pockets on the Rac1 surface that we believe are potential new binding sites. The Sites P1 and P2 are located in the region lining the switch regions I and II. These sites form the docking site for the GEFs that causes the opening of these sites large enough to attain volumes that can accommodate small molecule binders. In

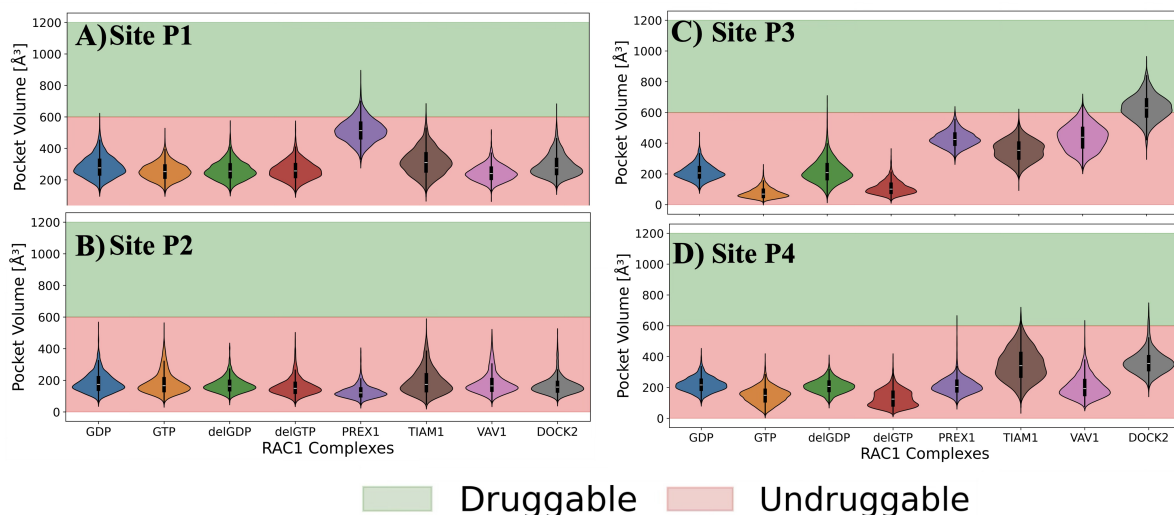


Figure 3: Four binding sites identified using POVME2. **A)** Shows the surface model of Rac1 protein with coloured spears illustrating the binding site identified. The following violin plots illustrate the distribution of the pocket volumes for each of these binding sites on the Rac1 surface monitored when Rac1 is complexed with either GEFs, GDP, GTP or in the *apo*-state. **B)** Site P1. It is the same site of the KRAS-G12C covalent inhibitors bind. **C).** Site P2 is found at the bottom of the switch I and II loops. **D).** Site P3 is found at the bottom of the switch I region. **E).** Site P4 is the only pocket that is not located close to switch I or II regions. The threshold for druggability of a pocket was set according to following references.^{35–37}

the absence of the GEFs, these sites are too small and hence not detected GEF-free Rac1(**Figure 2A**). Site P1 in Rac1 is similar to the one in K-RAS and H-RAS^{42,43} that is occupied by the covalent inhibitors in G12C RAS variants. Sites P1 and P2 are present at a crucial location that could modulate the dynamics of the switch regions. We hypothesise that one of the consequences of targeting Site P1 and P2 is, first, that the presence of a tight-binding molecule could prevent the recruitment of GEFs. Second, the molecules occupying these sites could act as molecular glues that stabilise or facilitate the interaction between Rac1 and GEFs. Both scenarios suggest that the modulation perturbs the GEF-mediated downstream signalling (**Figure3B**). Due to the differences in the binding volumes caused by different GEFs, we anticipate that this can be handled to develop molecules that could target Rac1 in GEF GEF-specific manner. It is also our belief that there exists a possibility to occlude GTP binding, albeit non-competitively, by preventing the dissociation of GEFs that is crucial for nucleotide exchange. Thereby prolonging the nucleotide-free state and preventing the downstream signalling. Site P2

is located below Site P1, and also found to marginal increase in the pocket volume in the presence of GEFs (**Figure 3C**). We propose that, owing to the proximity, this could serve as an extension for molecules binding to Site P1 (**Figure1**).

Away from the GEF binding site are located sites P3 and P4 (**Figure 2A**). Site P3 is located at the starting point of switch I. Targeting this site could modulate the switch I dynamics, which in turn, could have an influence on the traffic to the nucleotide-binding site as well as GEF binding. Residues lining the site P3 are known to interact more specifically with DHR-containing GEFs belonging to the DOCK family (*Add reference for PDB: 7DPA*). ELMO1 protein that facilitates the binding of DOCK family of GEFs is observed to interact with residues that form the site P4. Take taken together, we hypothesis that modulation of site P3 and P4 by small molecules could specifically target the pathologies arising from abnormalities related to DOCK family of GEFs.

Accelerating the pocket opening using Co-solvents

Transient pockets open in response to ligand binding suggesting an induced fit mechanism. However, in absence of known binders, co-solvents that structurally resemble drug-like fragments are used as chemical probes to accelerate opening of binding site that require ligand binding. In this work, we used three commonly used co-solvents with different hydrophobicities to further the nature of the binding site. PBD entry with accession number 3TH5 is the only X-ray crystal structure that is crystallised in the active-like conformation bound to GTP analogue, Guanylyl imidodiphosphate (GNP), with any regulatory protein. RAC1 conformation in this case shows that all GEF binding induced pockets are in the undruggable state (small pocket volumes). We performed microsecond timescale MD simulation to check the effectiveness of the chemical probes to open the transient pockets. We observe the three sites, P1, P3 and P4 visit conformations where the pocket volumes cross the threshold and are open enough to be druggable. However, system without co-solvent probes remain below the threshold (Figure4).

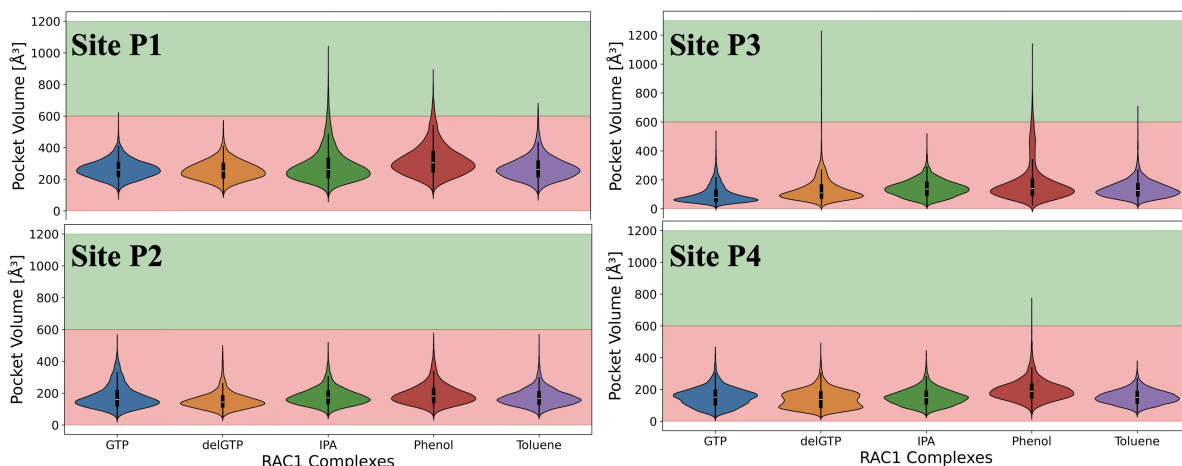


Figure 4: Accelerating transient pocket opening by Co-solvent MD simulations.

Role of switch region dynamics in formation of transient pockets.

In RAC1, like all other small GTPases, the switch regions should much higher conformational movements that were sensitive to nucleotide and GEF binding. The essential dynamics extracted from the dominate modes of motions suggested by the first principal component (XX%) also suggest that switch regions contribute to most of dynamics in the RAC1. Further, we embarked on understanding how the switch regions dynamics changes in response to GEF binding or nucleotide binding. The nucleotide-free and GEF-free states served to understand the changes inherent to RAC1 without of it binding partners.

Energetic contribution of residues involved in GEF binding

We hypothesise that most of the transient pockets in RAC1 and small GTPases in general open to accommodate the GEFs required for the nucleotide exchange. Therefore, we found it prudent to analyse the energetic contribution of the RAC1 residues that are involved in GEF binding, and their plausible role in binding small molecules. This can enable use to understand the nature of protein-ligand interactions the small molecules are likely to exhibit. As describes earlier, there is dominant role of the switch regions in GEF binding that leads to dissociation of GDP, and association of GTP. This nucleotide exchange catalysed by the GEF necessitates their interaction with the switch regions,

thus modulate their conformation such that it facilitates this exchange. The site P1, P2 and P3 are located the site where GEFs dock to the RAC1 protein to carry out its role. We also observe that the pocket volumes of these sites increase in the GEF bound state indicating that these site are crucial for GEF binding.

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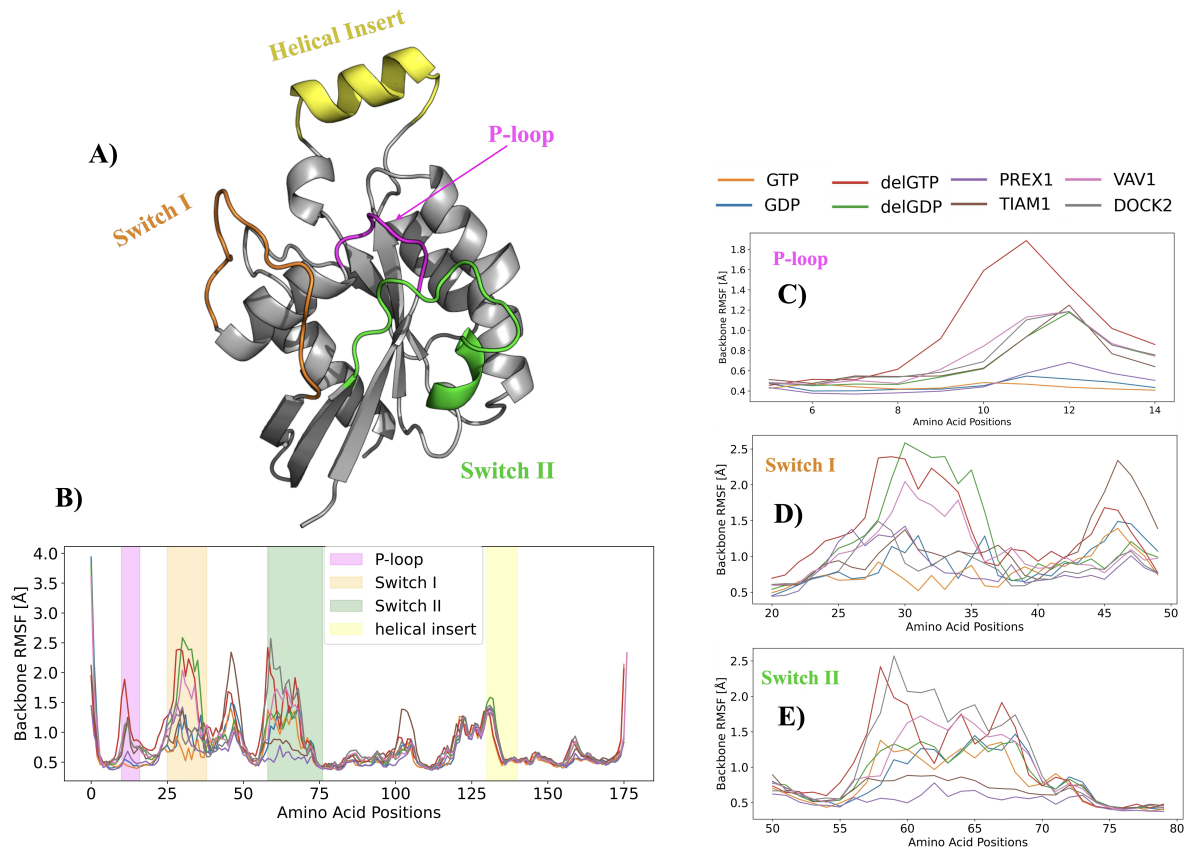


Figure 5: Rac1 domains and dynamics of each region. **A)** A Cartoon representation of human Rac1 (**PDB id:1HE1**). The P-loop region is depicted in the magenta colour. The Switch I region is depicted in Orange Colour, and the Green coloured region is the Switch II region. The helical region is shown in yellow colour. **B)** Atomic fluctuations of the backbone atoms are measured as root-mean-squared fluctuations (RMSF) for the human Rac1 domain. **C)** Backbone atom RMSF is shown as the expansion of the P-loop region. **D)** Backbone atom RMSF is shown as the expansion of the Switch I region. **E)** Backbone atom RMSF is shown as the expansion of the Switch II region. **F)** The normalised distribution of the root-mean-squared deviation of the backbone atoms of the P-loop. **G)** The normalised distribution of the root-mean-squared deviation of the backbone atoms of the Switch I region. **H)** The normalised distribution of the root-mean-squared deviation of the backbone atoms of the Switch II region. **delGDP** and **delGTP** are systems created by deleting the nucleotides from the Rac1 binding pocket to create corresponding *apo* system.

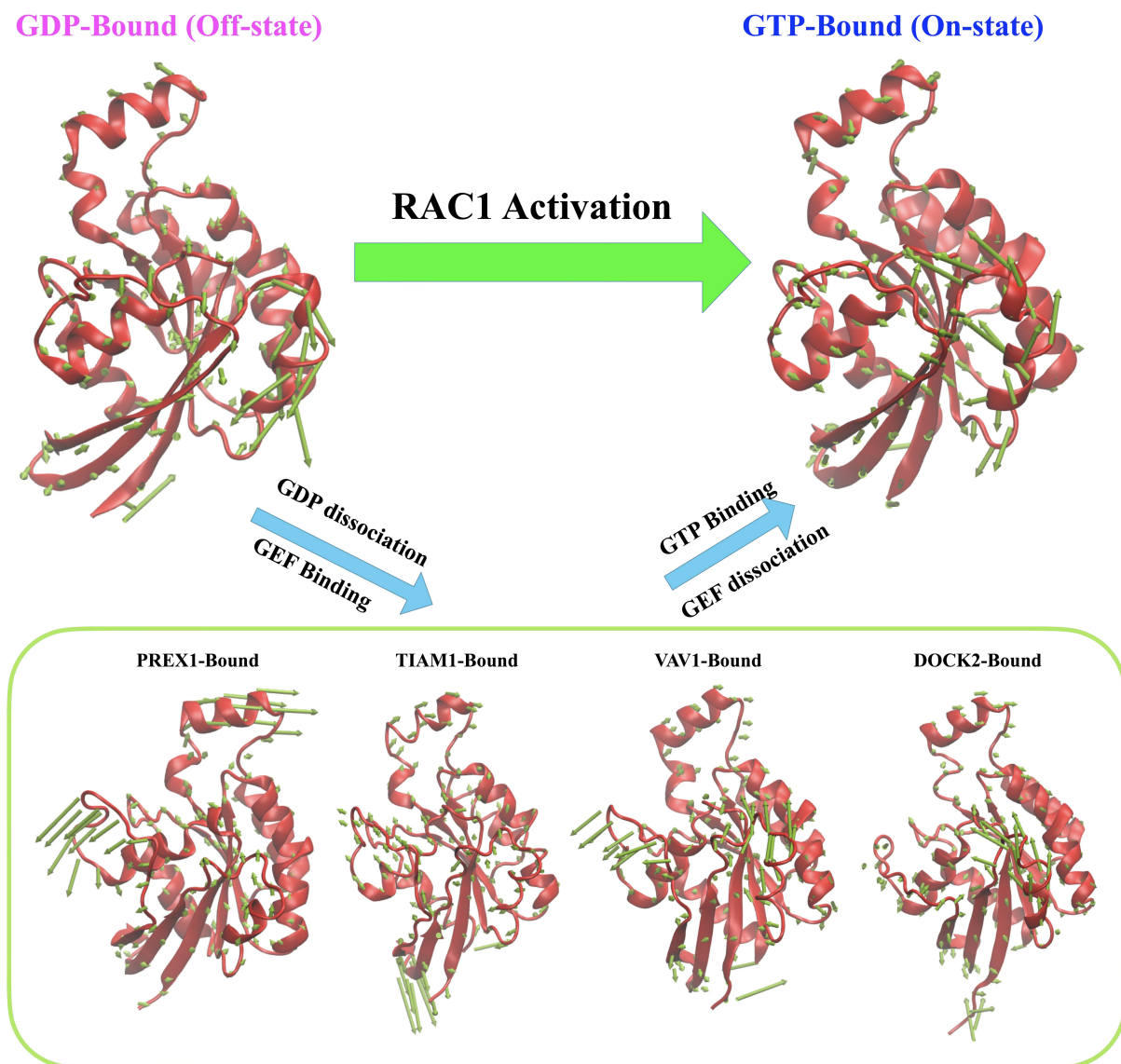


Figure 6: Essential Dynamics of RAC1 proteins to explain the opening and closing of the transient pockets. (TO DO: color code domains)

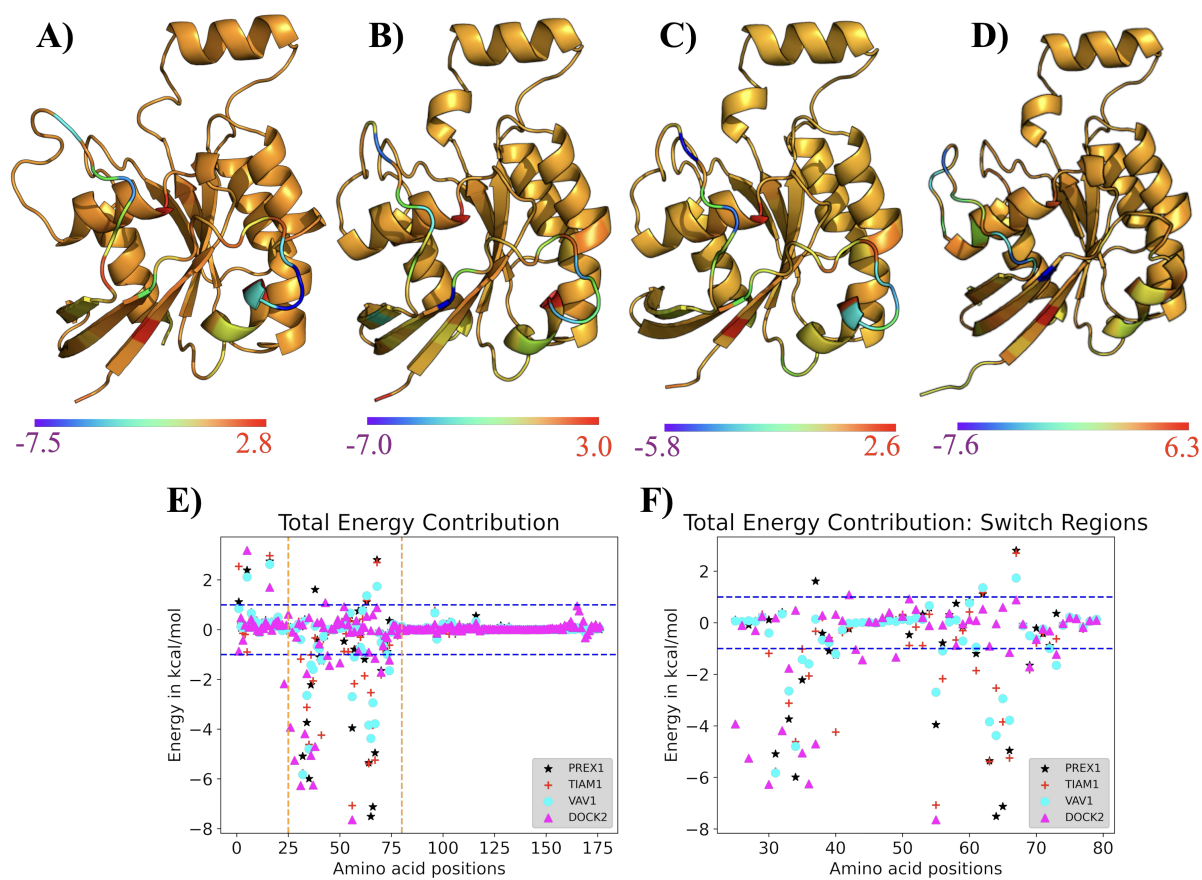


Figure 7: Per Residue contribution from Rac1 protein towards GEF binding.